When Satellite and Street Views Disagree: Cross-View Complementarity in Disaster Damage Assessment

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Abstract

Satellite and street-view imagery see different parts of a damaged building: roofs from above, facades and debris from the ground. Average accuracy hides whether fusion methods use this complementary evidence. We study the *conflict cases*, the samples on which independently trained single-view models disagree, as a diagnostic subset. On paired collections from the 2025 Eaton wildfire and Hurricanes Ian and Milton, conflicts make up 9.6–33.2% of the test data. Fusion improves conflict accuracy over either view, yet an oracle that selects the correct single view per conflict remains 0.19–0.32 above the best evaluated fusion method. A linear fusion gate over building-visibility features, calibrated confidences, and disagreement recovers part of this headroom where ground photographs are aimed at the building: on Eaton it beats end-to-end fusion by +0.072 conflict accuracy ($p < 10^{-4}$) and calibrated averaging by +0.051 ($p = 0.0001$); on the panoramic hurricane datasets it is indistinguishable from end-to-end fusion. Cropping panoramas toward the building raises the conflict gain of fusion, whereas random crops of the same size do not. The spatial density of conflicts, computed without labels, correlates with tile-level wildfire damage (Spearman $r = 0.615$, 25 tiles). Conflict cases thus show how much complementary evidence fusion uses, which view carries it under each hazard and capture setting, and where current methods fall short.

CCS Concepts

• **Computing methodologies** → **Computer vision**; *Machine learning*; • **Information systems** → *Geographic information systems.*

Keywords

cross-view complementarity, street-view imagery, satellite imagery, disaster damage assessment, cross-view fusion, GeoAI

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1 Introduction

Building-level damage assessment after a disaster increasingly draws on two image collections at once. Satellite imagery covers a whole region within hours and is the basis of standardized damage mapping [2], but it mainly shows roofs. Ground-level imagery, whether crowdsourced street-view panoramas or official inspection photographs, shows the facades and debris that overhead sensors cannot fully detect [11, 13]. The views are complementary: a facade can be gutted under an unbroken roof. Methods that pair them serve geolocalization and damage mapping [5, 6, 12], multimodal arbitration [10], and agentic assessment pipelines [8, 9], and usually report that fusion improves average accuracy over either view alone.

Average accuracy says little about whether this evidence is used, because for most buildings the two views agree. The informative records are the *conflict cases*, the samples on which independently trained single-view models disagree: at least one view carries evidence the other lacks, whether damage is visible only from one side, vegetation hides the building, or the views are misaligned. Conflicts are thus a focused subset for asking how much complementary evidence fusion recovers, which view carries it under a given hazard and capture setting, and how the views can be combined.

This study examines cases in which damage assessments derived from satellite imagery conflict with those derived from street-view imagery to address four research questions. (1) How much complementary headroom do fusion methods leave on conflict cases? We compare them with an oracle that selects whichever single view is correct. (2) Can a simple, readable gate recover part of it from cues about whether the ground image actually shows the building? (3) Does aiming the ground view at the building change the value of fusion? We vary framing *within* a dataset by controlled cropping. (4) Does the spatial density of conflicts, which needs no labels once the models are trained, track damage (Figure 1)? Code and protocols: https://github.com/rayford295/CrossViewGate.

2 Data and Protocol

We use three paired street/overhead collections spanning two hazards and two ground-view *observation regimes*: property-centric photographs aimed at a building, and panoramas that are not.

Eaton wildfire (Altadena, CA; property-centric). Building-level CAL FIRE Damage Inspection (DINS) photographs taken after

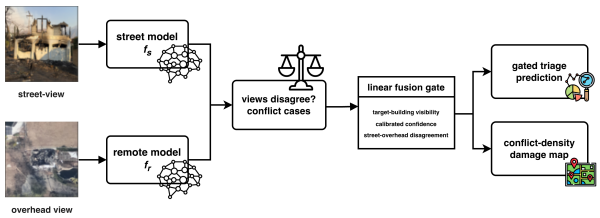


Figure 1: Overview. Single-view models define the conflict cases; a linear fusion gate decides how much weight each view receives per sample (the implemented gate also mixes in the end-to-end fusion model, omitted here); conflict density is a label-free damage indicator (Eaton-fire example).

Table 1: Test-set results (five-seed means; accuracy std ≤ 0.03). Conflict accuracy is computed on each seed's conflict subset. Closure follows Eq. (1) applied to the displayed means; the oracle reaches 0.961 / 0.896 / 0.964 conflict accuracy, i.e. 0.475 / 0.367 / 0.407 above the better single view.

	Eaton wildfire (conflict rate 9.6%)			Hurricane Ian (33.2%)			Hurricane Milton (25.7%)		
Method	Acc.	Conflict acc.	Closure	Acc.	Conflict acc.	Closure	Acc.	Conflict acc.	Closure
street_only	0.931	0.486	—	0.703	0.529	—	0.759	0.557	—
remote_only	0.929	0.475	—	0.648	0.367	—	0.721	0.407	—
concat	0.930	0.674	0.40	0.709	0.588	0.16	0.766	0.622	0.16
crossview (end-to-end fusion)	0.940	0.699	0.45	**0.735**	**0.624**	**0.26**	0.776	**0.649**	**0.23**
calibrated probability averaging	0.954	0.732	0.52	0.731	0.617	0.24	0.776	0.626	0.17
fusion gate (linear)	**0.959**	**0.768**	**0.59**	0.731	0.618	0.24	**0.777**	0.631	0.18

the January 2025 fire, paired by location with post-event Very-High-Resolution (VHR) overhead tiles. The six-level DINS scale is mapped to three ordinal classes (no/trace damage, damaged–repairable, destroyed). The split has 6.5k/1,966/1,984 train/val/test pairs. **Hurricane Ian (panoramic).** The CVIAN pairing released with CVDisaster [6]: Mapillary 360° panoramas over Sanibel Island, 27 cm post-event satellite imagery, and three-level damage-perception labels, with 3,248/573/300 pairs. **Hurricane Milton (panoramic).** A 360° street-view/satellite pairing over Florida after the October 2024 hurricane, built in the same way; 1,740/307/254 pairs.

Splits and leakage. Eaton is split by property and tile, so all photographs of one structure fall in one partition. For Ian, the headline numbers keep the released split for comparability with CVDisaster. However, an audit against the official CVIAN position file found that 299 out of the 300 test samples share a Mapillary sequence with the training data, so we also evaluate a repaired protocol with 0.005° spatial blocks and a 25 m buffer. Under it, macro-F1 drops by 0.05–0.10 for every model, so index-based splits overstate spatial generalization, although the single-view ranking (street above satellite) is unchanged. The Ian results in Table 1 should therefore be interpreted as in-distribution results, not as the evidence of spatial transfer.

3 Method

Base models and conflict cases. For each dataset, we trained four models under one protocol: *street_only* and *remote_only* (ResNet-18 [3] with a linear head), *concat* (both encoders with an MLP head), and *crossview* (both encoders with a learned interaction head), which serves as the end-to-end fusion reference. Training uses class-balanced cross-entropy, 224-pixel inputs, AdamW, up to 15 epochs with early stopping on validation macro-F1, and five seeds. Writing $s(x)$ and $r(x)$ for the two single-view predictions, the **conflict subset** is $C = \{x : s(x) \neq r(x)\}$. Conflict rates are 9.6% on Eaton, 33.2% on Ian, and 25.7% on Milton.

Oracle single-view gap. The oracle counts a conflict sample as correct if *either* single view is correct. For a method m we report its conflict accuracy $\mathrm{acc}_C(m)$ and its **oracle-gap closure**,

$$\mathrm{closure}(m) = \frac{\mathrm{acc}_C(m) - \mathrm{acc}_C(\text{best single})}{\mathrm{acc}_C(\text{oracle}) - \mathrm{acc}_C(\text{best single})}, \tag{1}$$

where "best single" is the single-view model with the higher mean conflict accuracy. The oracle only chooses *which* of two fixed models to use, so it measures the headroom of hard view selection; it is not an upper bound for probability mixtures, which can in principle exceed it.

Calibration. Since miscalibration can distort averaging [1], we fit one temperature per view on validation predictions and evaluated every training-free fusion baseline on calibrated probabilities; the argmax, and hence the conflict subset, is unchanged.

Visibility-conditioned fusion gate. A frozen SegFormer-B0 [7] trained on ADE20K gives four building-visibility features per street image: the building pixel ratio, the same ratio inside a centered crop of half the image size, their difference (a centering signal), and the normalized distance of the building centroid from the image center. These features describe visible built structures in general, not the specific target building. Seven more features describe the predictions themselves: calibrated confidence and entropy for each view, the confidence gap, the Jensen–Shannon divergence between the two calibrated distributions, and a disagreement flag. Given the eleven z-scored features z, the gate is a mixture of experts [4] with

linear logits,

$$p(y \mid x) = \sum_k w_k(z)\, p_k(y \mid x), \qquad w(z) = \mathrm{softmax}(Wz + b), \quad (2)$$

over $k \in \{\text{street}, \text{remote}, \text{crossview}\}$, fit on the validation split by minimizing the negative log-likelihood of the mixture. A *two-expert* variant (street and remote only) has a single logit, so each coefficient reads directly as a shift of weight toward one view. A small MLP in place of the linear logits did not help on Eaton.

Field-of-view intervention. Each 1024×512 panorama is cropped to a 256×256 window with a 90° horizontal field of view on the horizon band. In the **building-centered** variant the window is aimed at the circular-mean azimuth of the SegFormer building mask. In the **random** variant it is aimed at a reproducible uniformly random azimuth. The variants differ only in where the window points. We retrained *street_only* and *crossview* on each variant (three seeds, three-epoch schedule, also used for the original-panorama reference), reuse the remote models, and compare the gain of fusion on conflicts.

Conflict density. We averaged test-set predictions over seeds, aggregate them to geographic tiles, and computed per tile the hard conflict density (fraction of samples on which the two argmax predictions disagree) and, as a control, the mean single-view predictive entropy. Both are rank-correlated (Spearman) with the tile's mean ordinal damage label.

4 Results

Overall results and the oracle gap. For paired comparisons, we averaged each test sample's correctness difference over the conflict seeds and applied a two-sided sign-flip permutation test (20,000 permutations). Pooled differences weight samples rather than seeds, so they differ slightly from Table 1. First, end-to-end fusion beats the street model on conflicts on every dataset (Eaton +0.246, $p < 10^{-4}$; Ian +0.078, $p = 0.013$; Milton +0.141, $p = 0.006$). Second, once the single-view models are converged and calibrated, probability averaging is a strong baseline: on Eaton it exceeds end-to-end fusion in both overall accuracy (0.954 vs. 0.940) and conflict accuracy (0.732 vs. 0.699). Third, the oracle headroom dwarfs these differences: picking the right single view per sample would add 0.475 / 0.367 / 0.407 conflict accuracy over the better single view, and still 0.193 / 0.272 / 0.315 over the best evaluated fusion method.

The fusion gate. On the property-centric dataset the gate gains +0.072 conflict accuracy over end-to-end fusion ($p < 10^{-4}$) and +0.018 on the full test set ($p < 10^{-4}$); against calibrated averaging, the strongest baseline, it gains +0.051 on conflicts (95% bootstrap CI $[+0.026, +0.077]$, $p = 0.0001$). Its closure of 0.59 is the highest in Table 1. On the panoramic datasets the street views rarely show a building prominently (mean building pixel ratio 0.027 on Ian vs. 0.119 on Eaton), so the visibility features carry little signal, and the gate is statistically indistinguishable from end-to-end fusion (pooled differences +0.005, $p = 0.70$ on Ian; −0.015, $p = 0.57$ on Milton). How much of the Eaton gain comes from visibility-based weighting rather than from adding a third expert? The two-expert gate, which can only reweight street and remote, reaches 0.734 conflict accuracy, on par with calibrated averaging (0.732) and above

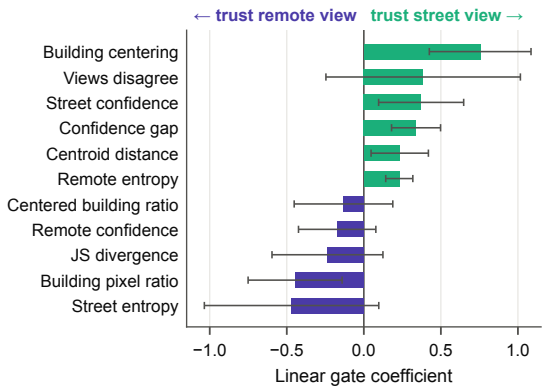


Figure 2: Coefficients of the two-expert (street vs. remote) linear gate on Eaton (mean ± std over five seeds). Positive values shift weight toward the street view; bars crossing zero have no stable sign across seeds.

Table 2: Field-of-view intervention (three seeds, three-epoch schedule; remote models unchanged). Conflict gain = crossview − best single view on each variant's own conflict subset.

Dataset	Street-view variant	Street acc.	Conflict gain
Ian	original panorama	0.709	+0.064
Ian	building-centered crop	0.648	**+0.109**
Ian	random crop (control)	0.574	+0.042
Milton	original panorama	0.740	+0.067
Milton	building-centered crop	0.693	**+0.149**
Milton	random crop (control)	0.654	+0.055

end-to-end fusion (+0.034, $p = 0.047$). Adding crossview as a third expert lifts this to 0.768, so the headline gain combines both effects. The two-expert coefficients (Figure 2) are the interpretable part. The largest weight is building centering ($+0.75 \pm 0.33$), followed by the confidence gap and street confidence, while the raw building pixel ratio is negative (-0.45 ± 0.31). Weight on the street view thus rises when the photograph is confident and the building sits in the middle of the frame, not merely when much of it is built-up area.

Field-of-view intervention. Building-centered cropping raises the conflict gain of fusion on both hurricanes (Table 2): from 0.064 to 0.109 on Ian and from 0.067 to 0.149 on Milton. Random crops with identical geometry do not (0.042 and 0.055), and the building-minus-random contrast is positive in all six dataset–seed pairs. Cropping *lowers* street-only accuracy yet *raises* the benefit of fusion, so the gain does not come from a stronger street model, and the random control argues against a cropping artifact. This supports building-oriented framing as a driver of the regime effect, although a fixed-sample test with target-building annotations would be stronger.

Conflict density as a spatial damage signal. On the wildfire, the conflict density of an overhead tile tracks its damage (Spearman $r = 0.615$, $p = 0.001$ over 25 tiles; $r = 0.60$ on the 0.01° grid of Figure 3). The uncertainty control behaves very differently: it *anti-correlates* with damage ($r = -0.37$ to -0.52), because parcels burned to the ground are easy, confident classifications. Damage on Eaton

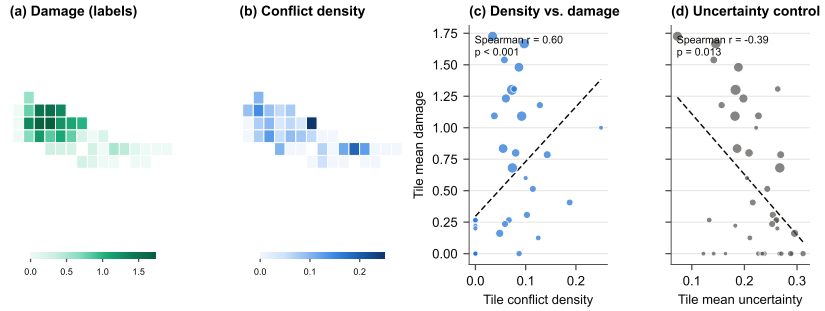


Figure 3: Conflict density as a label-free damage indicator (Eaton wildfire, 0.01° grid). (a) Labeled tile damage. (b) Tile conflict density from model predictions. (c) Their rank correlation. (d) The single-view uncertainty control, which anti-correlates with damage. Point size reflects tile sample count; p-values ignore spatial dependence.

is spatially autocorrelated (Moran's $I = 0.27$), so the nominal p-values overstate the evidence. On Milton the signal is weaker and not significant once the models converge ($r = 0.289$, $p = 0.14$), because better single-view models disagree less.

5 Discussion and Conclusion

Conflict cases make the value of complementary views measurable. The oracle headroom, a restricted choice between two existing models, exceeds every fusion gain we observed, so current fusion leaves most view-specific evidence unused. Which view carries it depends on hazard and capture setting. On property-centric wildfire photographs, visibility cues identify when the street view is decisive, and the gate, one frozen segmentation pass plus a linear fit, is cheap and auditable: weight the street view when it is confident and the building is centered. On panoramic hurricane imagery the building is rarely prominent, and calibrated averaging or end-to-end fusion suffices, and framing ground photographs on structures raises the value of fusion.

Limitations. Hazard and capture regime are confounded across datasets, and the property-centric regime rests on one wildfire. Milton is small (254 test pairs); the released Ian split has sequence overlap and the repaired protocol only six test blocks. The permutation tests ignore dependence between nearby properties. We measure prediction complementarity, not which component was damaged; conflicts may also stem from occlusion, misalignment, or label ambiguity. The density signal needs spatially aware inference and replication.

Conclusion. Where satellite and street views disagree, symmetric fusion leaves substantial complementary evidence unused. A visibility-conditioned gate recovers part of it for building-aimed photographs, and spatially aggregated disagreement is a label-free damage indicator. Mining conflicts turns "does fusion help?" into "which view carries the evidence, where, and why?".

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References

[1] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. On Calibration of Modern Neural Networks. In *Proceedings of the 34th International Conference on Machine Learning*. PMLR, 1321–1330.

[2] Ritwik Gupta, Bryce Goodman, Nirav Patel, Ricky Hosfelt, Sandra Sajeev, Eric Heim, Jigar Doshi, Keane Lucas, Howie Choset, and Matthew Gaston. 2019. Creating xBD: A Dataset for Assessing Building Damage from Satellite Imagery. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*. 10–17.

[3] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep Residual Learning for Image Recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 770–778.

[4] Robert A. Jacobs, Michael I. Jordan, Steven J. Nowlan, and Geoffrey E. Hinton. 1991. Adaptive Mixtures of Local Experts. *Neural Computation* 3, 1 (1991), 79–87.

[5] Hao Li, Fabian Deuser, Wenping Yin, Steffen Knoblauch, Wufan Zhao, Filip Biljecki, Yong Xue, and Wei Huang. 2026. Towards Generative Location Awareness for Disaster Response: A Probabilistic Cross-View Geolocalization Approach. *ISPRS Journal of Photogrammetry and Remote Sensing* 237 (2026), 130–145.

[6] Hao Li, Fabian Deuser, Wenping Yin, Xuanshu Luo, Paul Walther, Gengchen Mai, Wei Huang, and Martin Werner. 2025. Cross-View Geolocalization and Disaster Mapping with Street-View and VHR Satellite Imagery: A Case Study of Hurricane IAN. *ISPRS Journal of Photogrammetry and Remote Sensing* 220 (2025), 841–854.

[7] Enze Xie, Wenhai Wang, Zhiding Yu, Anima Anandkumar, Jose M Alvarez, and Ping Luo. 2021. SegFormer: Simple and Efficient Design for Semantic Segmentation with Transformers. In *Advances in Neural Information Processing Systems*, Vol. 34. 12077–12090.

[8] Yifan Yang, Wenjing Gong, Kaili Zhang, Lei Zou, Zhengzhong Tu, Hao Li, Zongrong Li, and Xinyue Ye. 2026. RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery. *arXiv preprint arXiv:2606.21819* (2026).

[9] Yifan Yang and Lei Zou. 2026. RAPIDMap: Rapid Multi-Agent Pipeline for Interpretable Disaster Mapping from Satellite and Street-view Imagery. *arXiv preprint arXiv:2609.00046* (2026).

[10] Yifan Yang, Lei Zou, Wenjing Gong, Kani Fu, Zongrong Li, Siqin Wang, Bing Zhou, Heng Cai, and Hao Tian. 2026. DamageArbiter: A CLIP-Enhanced Multimodal Arbitration Framework for Hurricane Damage Assessment from Street-View Imagery. *arXiv preprint arXiv:2603.14837* (2026).

[11] Yifan Yang, Lei Zou, Bing Zhou, Daoyang Li, Binbin Lin, Joynal Abedin, and Mingzheng Yang. 2025. Hyperlocal Disaster Damage Assessment Using Bi-Temporal Street-View Imagery and Pre-Trained Vision Models. *Computers, Environment and Urban Systems* 121 (2025), 102335. doi:10.1016/j.compenvurbsys.2025.102335

[12] Wenping Yin, Fabian Deuser, Ziqi Liu, Jiaqi Wei, Xuanshu Luo, Martin Werner, Hao Li, and Yong Xue. 2026. Triple-Objective Cross-View Geolocalization of Disaster-Related VGI: The Case of Hurricane Ian. *International Journal of Geographical Information Science* 40, 1 (2026), 194–216.

[13] Wei Zhai and Zhong-Ren Peng. 2020. Damage Assessment Using Google Street View: Evidence from Hurricane Michael in Mexico Beach, Florida. *Applied Geography* 123 (2020), 102252.